SLE-4: Architecture Decision Record (ADR)

Course: 02AML204 –Introduction to Artificial Intelligence

PRN: 25UET106 | Name: NihalAmbi | Division: B | Date: 12 October 2026

PARTA: Architecture Decision Record (ADR)

Title: Use Breadth-First Search (BFS) asthe Main Solver for Maze Path-Finding

1.Context

1. The project comparesBFS and Depth-Limited DFS for maze path-finding.
2. The maze isasolvable grid with the start at the top-left and the goalat the bottom-right.
3. The search usesBFS and depth-limited DFS with alimit of 1,000.
4. The SLE-2 repository records tests on 20×20, 40×40 and 70×70 mazes. It measures execution time and expanded nodes.
5. The SLE-3 repository explainsthe system architecture using the C4 model.

2. Decision

I chose BFS as the main solver when the shortest path is required. DFS remains in the project for comparison. The SLE-2 README states that BFS found the true shortest path in the tested mazes, while DFS did not.

3. Alternatives Considered

Depth-Limited DFS (limit1000):used asthe comparisonalgorithm. Inthereportedtests, itwas fasterandexpanded fewer nodes, but it did not find the shortest path.

BFS: selected asthe main solver because it returned the shortest path in the reported test cases.

4. Rationale / Justification

The values below are taken from the SLE-2 repository README. The README notes that exact figurescan vary between runsbecause mazesare randomly generated.

Maze size	BFS	DFS steps	BFS nodes	DFS nodes
20 × 20	steps 38	72	295	155
40×40	78	166	1,212	674
70 × 70	138	390	3,739	2,421

BFS returned shorter pathsin allthree reported cases.

DFS expanded fewer nodes in the recorded tests. This depends on the maze wall layout and the fixed neighbour order (up, down, left, right); it doesnot mean DFS isalwaysbetter.

The SLE-3 architecture lists Maze Generator, Search Engine, Path Reconstructor, Output & Metrics, and Experiment & Profiling.The Search Engine includesFrontier, Visited Set, GoalTest, Parent Map, BFS and DFS.

5. Consequences

Po sit iv e :

BFS iscomplete and optimalon an unweighted maze, asstated in the SLE-2 README.

The project can compare path length, expanded nodesand execution time for BFS and DFS.

Negative /trade-off:

BFS expanded more nodesthan DFS in the reported test cases.

The SLE-2 README givesBFS time and space complexity asO(b^d).

6. References

SLE-1/IAI:
https://github.com/nihalambi32/IAI SLE-2:
https://github.com/nihalambi32/SLE-2
SLE-3:
https://github.com/nihalambi32/SLE-3

PARTB: Stepwise Summary of SLE-1, SLE-2 and SLE-3

SLE-1:StudentAIAssistant(IAIrepository)

Concept: A simple rule-based conversational program written in Python. It responds to specific student-related keywords with prepared answers. It is not described in the README asatrained machine-learning model.

1. Open the program: the main function isstudent_agent() in ai_agent.py.
2. The assistant displaysitstitle and tellsthe user to type “bye” to exit.
3. The user typesamessage.The program convertsthe message to lowercase.
4. The program checks the message for known words or phrases, such as hello/hi, python, DSA/data structure, exam, college or study.
5. When aknown keyword isfound, it printsthe matching prepared response.
6. When the user types“bye”, the program printsagoodbye message and stops.
7. If no known keyword matches, the program printsthat it doesnot have an answer.
8. GitHub Copilot was used to generate or modify selected code sections. The README says suggestions were reviewed, tested and modified where needed; the contribution log recordsthe selected contributions.

Technologies listed in the README: Python 3.12+, Visual Studio Code, Git, GitHub and GitHub Copilot.The repository containsai_agent.py, README (1).md and Contribution_Log(2).md.

SLE-2: BFS vs DFS Maze Profiling

9. Generate asolvable grid maze with astart and goal.
10. Run BFS and Depth-Limited DFS to search for apath.
11. Repeat the experimentson 20×20, 40×40 and 70×70 mazes.
12. Compare the path length, expanded nodes and execution time. The repository mentions time.perf_counter and py-spy for profiling.
13. Reported result: BFS found shorter paths; DFS expanded fewer nodesand wasfaster in the recorded tests.

SLE-3: C4 Architecture

14. Use the C4 modelto describe the structure of the BFS/DFS maze project.
15. The four C4 levelsare Context, Container, Component and Code.
16. The documented components are Maze Generator, Search Engine, Path Reconstructor, Output & Metrics, and Experiment & Profiling.
17. The Search Engine includesFrontier, Visited Set, GoalTest, Parent Map, BFS and DFS.

PARTC: AI Contribution Note

18. The AIcoding toolused in SLE-1wasGitHub Copilot.
19. The IAIREADME saysCopilot helped generate or modify selected partsof the Python program.
20. The contribution log records suggestions for the student_agent() structure, the while loop, bye exit, greeting, Python, DSA, study advice and default response.
21. The student reviewed, tested or modified the suggestions rather than accepting the whole program without understanding it.